Classifying Power Plant Fuel Types via Emissions Fingerprinting

A Supervised Gaussian Naive Bayes Analysis of 2007 EPA eGRID Data

Alexander Elamthuruthy

1 Summary

Identifying the generation source of power facilities is critical for emissions monitoring, environmental compliance, and regional grid modeling. This study evaluates whether a power plant's primary fossil fuel type (Coal, Natural Gas, Oil, or Other Fossil) can be classified strictly through its annual output of four core air emissions: CO_2, SO_2, NO_x, and CH_4.

Using plant-level records from the EPA's Emissions & Generation Resource Integrated Database (eGRID2007), raw skewed emissions features were normalized via a $\log(1 + x)$ transformation to align with Gaussian distributional assumptions. A Gaussian Naive Bayes model trained on a 70/30 split achieved an overall test accuracy of 77.6%. The classifier demonstrated strong predictive capability in isolating coal-fired generation due to pronounced SO_2 profiles, while highlighting the vulnerability of probabilistic classifiers to severe real-world class imbalances when distinguishing minority fuel categories.

2 Description of the Data

This data set belongs to the Agriculture/Environment/Ecology/Climate data area. Specifically, it provides data about the environmental characteristics for most of the electric power generated in the United states for the year 2007. The data includes emissions, emission rates, generation, heat input, resource mix, and may other attributes.

The objective of this study is to determine the extent to which a power plant's primary fuel source can be identified solely through its emission fingerprint. This research seeks to quantify the predictive power of NO_x, SO_2, CO_2, and CH_4 outputs in distinguishing between the different fuel sources used at power plants.

Data for this analysis was sourced from the U.S. Environmental Protection Agency's (EPA) Emissions & Generation Resource Integrated Database (eGRID). Specifically, observations were extracted from the 2007 plant-level dataset (PLNT07), which compiles annual environmental, operational, and generation metrics across power facilities operating within the United States.

For this study we will be using the variables

- **PLFUELCT**: Primary Fossil Fuel Category, the categorical target variable (Coal, Gas, Oil, or Other Fossil).
- **PLCO2AN**: Annual CO_2 Emissions: Plant Annual carbon dioxide emitted (tons).
- **PLSO2AN**: Annual SO_2 Emissions: Plant Annual sulfur dioxide emitted (tons).
- **PLNOXAN**: Annual NO_x Emissions: Plant Annual nitrogen oxides emitted (tons).
- **PLCH4AN**: Annual CH_4 Emissions: Plant Annual methane emitted (tons).

3 Analysis & Discussion

To find the predictive power of the emissions a Naive Bayes model was used for classification. The response variable was PLFUELCT, the primary fossil fuel used at the plant that had a value of either Coal, Gas, Oil, or Other Fossil Fuel.

The Predictors were:

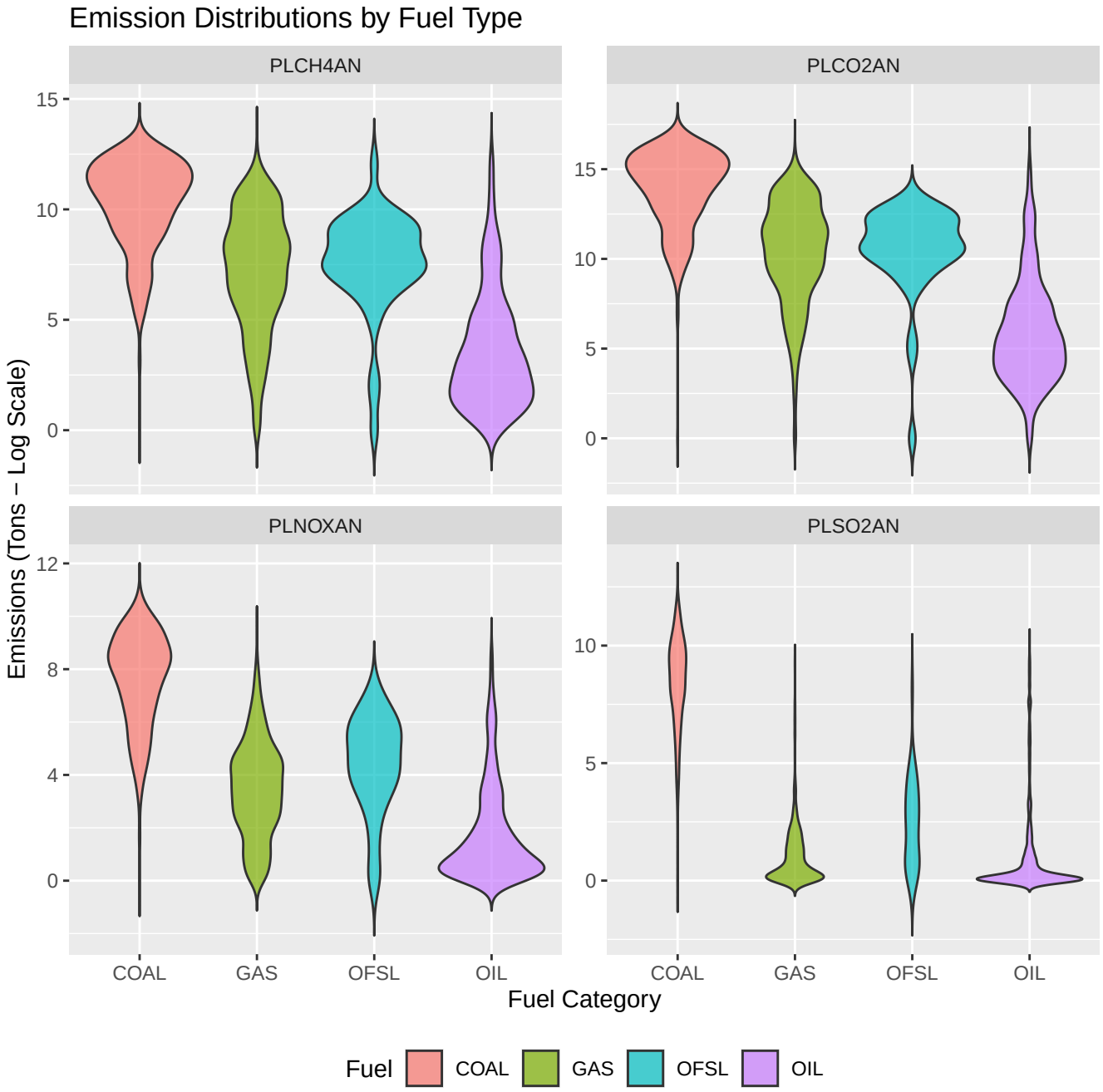
- **PLCO2AN**: Annual CO_2 Emissions
- **PLSO2AN**: Annual SO_2 Emissions
- **PLNOXAN**: Annual NO_x Emissions
- **PLCH4AN**: Annual CH_4 Emissions

Table 1: Plant means by Fuel Category

PLFUELCT	Avg CO2 (Tons)	Avg SO2 (Tons)	Avg NOx (Tons)	Avg CH4 (Tons)
COAL	3782559	15851	5536	87768
GAS	292271	41.98	153.7	11933
OFSL	117629	112	219.5	9103
OIL	48807	240.9	94.67	3467

As evidenced by the summary statistics in Table 1, the raw emission means show a massive difference in scale. For example, the average CO_2 output for Coal plants is measured in the millions of tons, while the corresponding SO_2 and NO_x averages are

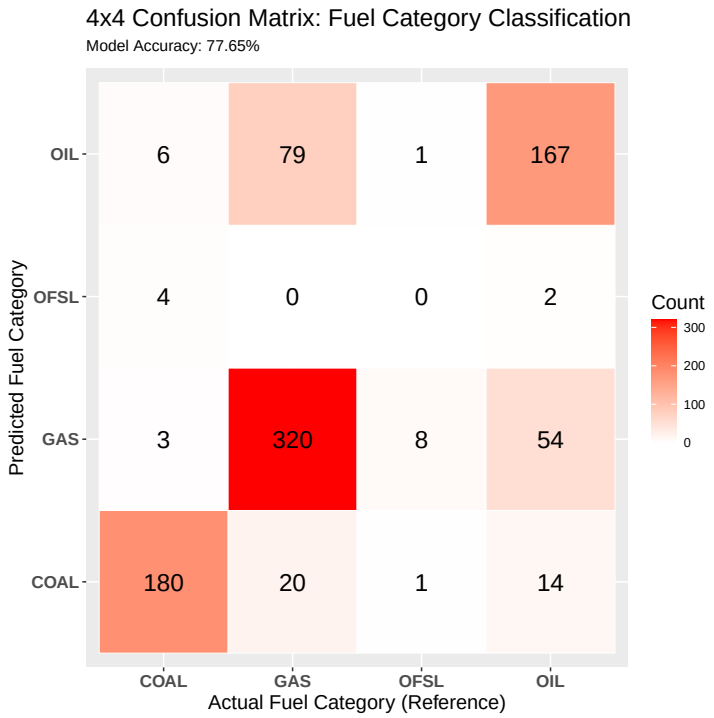
much smaller in comparison. This extreme variance creates a heavily right-skewed distribution that violates the fundamental assumptions of a Gaussian Naive Bayes classifier. To address this, we applied a log transformation ($log1p$) to all emission features.



Looking at the violin plot we can see the distribution of data values for all the emissions based on the fuel used at a plant. When looking at the plots we can predict that the model will be very accurate at identifying Coal due to the separation of coal from the other fuels in the SO_2 plot. We can also predict that the model may struggle with predicting oil due to the similar shape is has in all four of the plots

After cleaning the data and performing the log transformation, the data was split into a training set and a test set, the training set holding 70% of the data with the test set holding the other 30%.

After fitting a model with the training set, we used said model to get predictions of what the fossil fuel type should be based off of the emissions. The predicted values were then compared to the test set in a confusion matrix.



The model achieved an overall classification accuracy of 77.65%. While this figure suggests a generally effective model, a deeper examination of the confusion matrix reveals significant differences in performance across different fuel categories.

The model demonstrated its highest reliability in identifying Coal, strengthening the prediction that the difference in SO_2 compared to the other fuels would make it identifiable. However, the model struggled significantly with the remaining categories, most notably Other Fossil Fuel (OFSL), which yielded a 0% success rate.

This failure is primarily attributed to Representational Bias within the dataset. As shown in the matrix, the Gas category contains over 300 observations, whereas OFSL contains fewer than 15. In a Gaussian Naive Bayes framework, this imbalance causes the model to have a very poorly defined "bell curve" for rare classes, making it mathematically more likely to default to a more frequent class like Gas or Coal when emissions data is ambiguous.

Consequently, while the model serves as a tool for identifying high-emission coal facilities, its lack of sensitivity toward fuel types with similar emissions makes it an unreliable standalone predictor for the broader fuel mix.

4 Citations

U.S. Environmental Protection Agency (EPA). Emissions & Generation Resource Integrated Database (eGRID), 2007 Plant Data. Washington, DC: U.S. EPA.